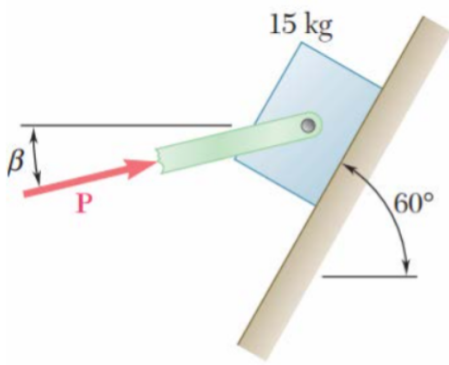


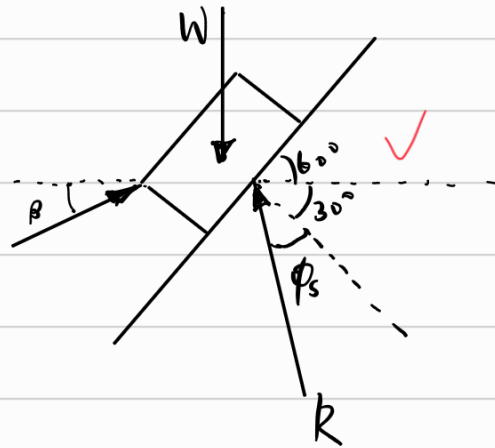
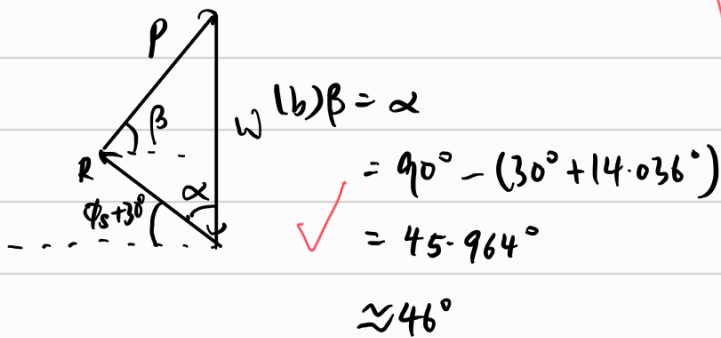
Grade: 5/5 Excellent!



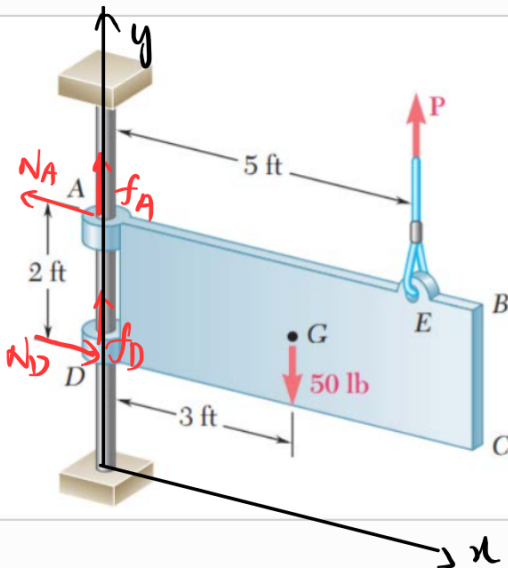
8.7 Knowing that the coefficient of friction between the 15-kg block and the incline is $\mu_s = 0.25$, determine (a) the smallest value of P required to maintain the block in equilibrium, (b) the corresponding value of β .

take $g = 9.81 \text{ m/s}^2$ $W = 15(9.81) = 147.15 \text{ N}$ $\tan \phi_s = \mu_s$ $\phi_s = \tan^{-1} \mu_s = 14.036^\circ$

smallest value of $P \Rightarrow P \perp R$ R - resultant force of F_s and N



(a) $P = 147.15 \sin 45.964^\circ$
 $= 105.79 \text{ N}$



8.30 The 50-lb plate $ABCD$ is attached at A and D to collars that can slide on the vertical rod. Knowing that the coefficient of static friction is 0.40 between both collars and the rod, determine whether the plate is in equilibrium in the position shown when the magnitude of the vertical force applied at E is (a) $P = 0$, (b) $P = 20 \text{ lb}$.

(a) $P = 0$

$\sum M_D = 0$ $N_A(2) - 3(50) = 0$ $N_A = 75 \text{ lb}$ $f_{A \max} = \mu_s N_A = 0.4(75) = 30 \text{ lb}$

$\sum F_x = 0$ $N_D = N_A = 75 \text{ lb}$

$\sum F_y = 0$ $f_A + f_D > 50 \text{ lb}$ $f_{D \max} > f_{A \max} = 30 \text{ lb}$

$f_A + f_D = 50 \text{ lb} < f_{A \max} + f_{D \max} = 60 \text{ lb}$

Hence, the plate is in equilibrium when vertical force P applied at E is $P = 0$.

$$(b) P = 2016 \quad \left\{ \begin{array}{l} \sum M_D = 0 \quad N_A(2) - 3(50) + 5(20) = 0 \quad N_A = 2516 \\ \sum F_x = 0 \quad N_D = N_A = 2516 \\ \sum F_y = 0 \quad f_A + f_D + 2016 - 5016 = 0 \end{array} \right.$$

$$\sum F_x = 0 \quad N_D = N_A = 2516$$

$$\sum F_y = 0 \quad f_A + f_D + 2016 - 5016 = 0$$

$$f_A + f_D = 3016$$

$$f_{Amax} = \mu_s N_A = 0.4(25) = 1016$$

$$f_{Dmax} = f_{Amax} = 1016$$

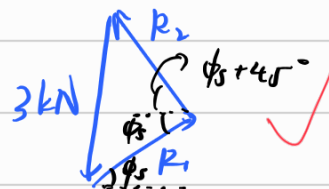
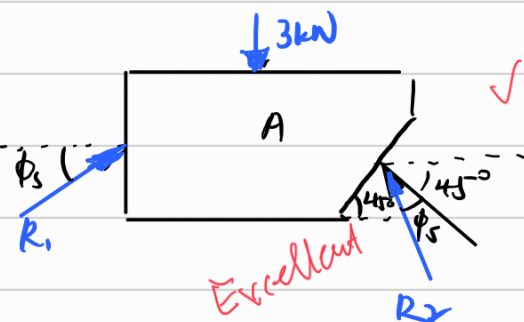
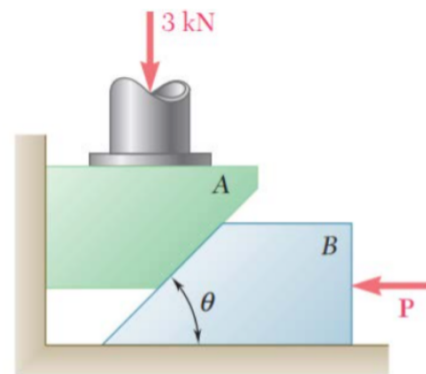
$$f_A + f_D > f_{Amax} + f_{Dmax}$$

Hence, the plate moves downwards.

8.54 Block A supports a pipe column and rests as shown on wedge B. Knowing that the coefficient of static friction at all surfaces of contact is 0.25 and that $\theta = 45^\circ$, determine the smallest force **P** for which equilibrium is maintained.

$$\phi_s = \tan^{-1} \mu_s = 14.036^\circ$$

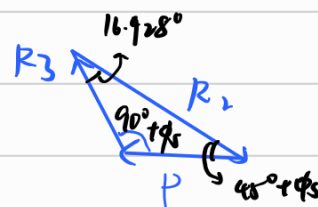
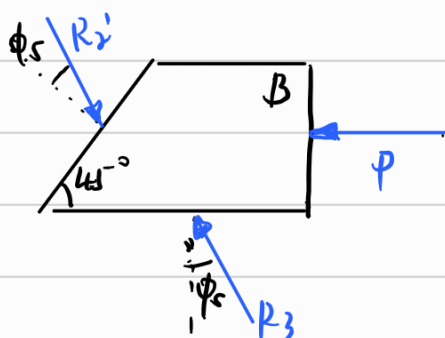
Consider Block A and pipe column as a system.



By sin rule,

$$\frac{3}{\sin(2\phi_s + 45^\circ)} = \frac{R_2}{\sin(90^\circ - \phi_s)}$$

$$R_2 = 3.042 \text{ kN}$$

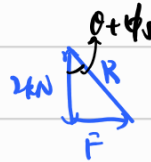
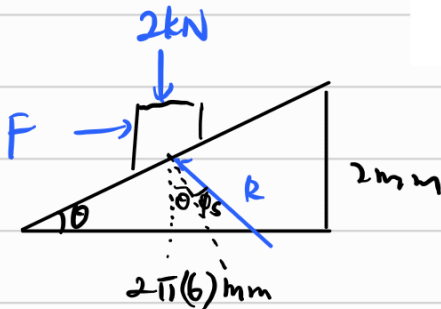


By sin rule,

$$\frac{R_2}{\sin(90^\circ + \phi_s)} = \frac{P}{\sin 16.928^\circ}$$

$$P = 0.913 \text{ kN}, \leftarrow$$

8.70 The ends of two fixed rods A and B are each made in the form of a single-threaded screw of mean radius 6 mm and pitch 2 mm. Rod A has a right-handed thread and rod B has a left-handed thread. The coefficient of static friction between the rods and the threaded sleeve is 0.12. Determine the magnitude of the couple that must be applied to the sleeve in order to draw the rods closer together.



$$\tan \theta = \frac{2}{2\pi(6)}$$

$$\theta = 3.037^\circ$$

$$\tan \phi_s = \mu_s$$

$$\phi_s = \tan^{-1} 0.12 = 6.843^\circ$$

$$\tan (\theta + \phi_s) = \frac{F}{2 \text{ kN}}$$

$$F = 2 \text{ kN} \tan (\theta + \phi_s)$$

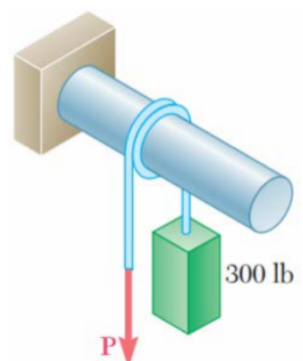
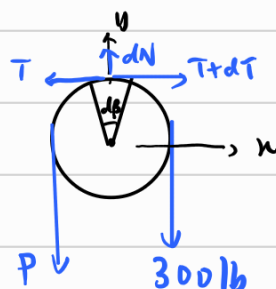
$$F = 0.348 \text{ kN}$$

$$\text{Magnitude of moment at A} = Fd = 348(0.006) = 2.088 \text{ N}\cdot\text{m}$$

same moment is required at B

$$\Rightarrow \text{Total magnitude of couple} = 2 \times 2.088 = 4.176 \text{ N}\cdot\text{m}$$

8.104 A 300-lb block is supported by a rope that is wrapped $1\frac{1}{2}$ times around a horizontal rod. Knowing that the coefficient of static friction between the rope and the rod is 0.15, determine the range of values of P for which equilibrium is maintained.



In equilibrium,

$$R_x = 0: T + dT - T = dF \Rightarrow dT = dF$$

$$R_y = 0: dN = (T + dT) \sin(d\beta/2) + T \sin(d\beta/2)$$

$$dN = T d\beta$$

$$\left(\sin\left(\frac{d\beta}{2}\right) \approx \frac{d\beta}{2} \text{ for small angle}\right)$$

$$|dF| \leq \mu_s dN$$

$$\Rightarrow |dT| \leq \mu T d\beta$$

$$\left| \int_P^{300} \frac{dT}{T} \right| \leq \mu \int_0^{3\pi} d\beta$$

$$\left| \ln \frac{300}{P} \right| \leq \mu \beta$$

$$-0.15(3\pi) \leq \ln \frac{300}{P} \leq 0.15(3\pi)$$

$$e^{-0.15(3\pi)} \leq \frac{300}{P} \leq e^{0.15(3\pi)}$$

$$P \geq \frac{300}{e^{0.15(3\pi)}} = 72.9716$$

$$P \leq \frac{300}{e^{-0.15(3\pi)}} = 1233.36216$$

$$\Rightarrow 1233.36216 \leq P \leq 72.9716$$